

How to Probe Sentence Embeddings in Low-Resource Languages: On Structural Design Choices for Probing Task Evaluation

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Abstract

Sentence encoders map sentences to real valued vectors for use in downstream applications. To peek into these representations—e.g., to increase interpretability of their results—probing tasks have been designed which query them for linguistic knowledge. However, designing probing tasks for lesser-resourced languages is tricky, because these often lack large-scale annotated data or (high-quality) dependency parsers as a prerequisite of probing task design in English. To investigate how to probe sentence embeddings in such cases, we investigate sensitivity of probing task results to structural design choices, conducting the first such large scale study. We show that design choices like size of the annotated probing dataset and type of classifier used for evaluation do (sometimes substantially) influence probing outcomes. We then probe embeddings in a multilingual setup with design choices that lie in a ‘stable region’, as we identify for English, and find that results on English do not transfer to other languages. Fairer and more comprehensive sentence-level probing evaluation should thus be carried out on multiple languages in the future.

1 Introduction

Sentence embeddings (a.k.a. sentence encoders) have become ubiquitous in NLP (Kiros et al., 2015; Conneau et al., 2017), extending the concept of word embeddings to the sentence level. In the context of recent efforts to open the black box of deep learning models and representations (Linzen et al., 2019), it has also become fashionable to *probe* sentence embeddings for the linguistic information signals they contain (Perone et al., 2018), as this may not be clear from their performances in downstream tasks. Such probes are linguistic micro tasks—like detecting the length of a sentence or its dependency tree depth—that have to be solved by

		classifier			
		LR	MLP	NB	RF
size	High	(A,B,C)	(A,B,C)	(C,A,B)	(C,B,A)
	Mid	(A,C,B)	(C,B,A)	(A,B,C)	(C,B,A)
	Low	(A,B,C)	(B,A,C)	(B,C,A)	(A,B,C)

Table 1: Schematic illustration of our concept of stability across two dimensions (classifier and training size). Here, three encoders, dubbed A,B,C, are ranked. The region of stability is given by those settings that support the majority ranking of encoders, which is $A \succ B \succ C$.

a classifier using given representations.

The majority of approaches for probing sentence embeddings target English, but recently some works have also addressed other languages such as Polish, Russian, or Spanish in a multi- and cross-lingual setup (Krasnowska-Kieraś and Wróblewska, 2019; Ravishankar et al., 2019). Motivations for considering a multi-lingual analysis include knowing whether findings from English transfer to other languages and determining a universal set of probing tasks that suits multiple languages, e.g., with richer morphology and freer word order.

Our work is also inspired by probing sentence encoders in multiple (particularly low-resource) languages. We are especially interested in the formal structure of probing task design in this context. Namely, when designing probing tasks for low-resource languages, some questions arise naturally that are less critical in English. One of them is the size of training data for probing tasks, as this training data typically needs to be (automatically or manually) annotated, an inherent obstacle in low-resource settings.¹

Thus, at first, we ask for the training data size

¹The main issue is that high-quality dependency parsers, as required for standard probing tasks, exist only for a handful of languages. E.g., UDPipe (Straka, 2018) is available for only about 100 languages, and performance scores for some of these are considerably below those of English (Straka, 2018).